

Safety Data Sheet

Lucantin® Red

Revision date : 2017/11/22

Version: 1.0

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(30041146/SDS_GEN_DO/EN)

1. Identification

Product identifier used on the label

Lucantin® Red

Recommended use of the chemical and restriction on use

Recommended use*: feed additive(s)

* The "Recommended use" identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:

BASF Dominicana S.A
Av. Winston Churchill
Acropolis Center Tower
8vo Piso. SPATIUM
Pinatini, 10148
Santo Domingo, República Dominicana
Telephone: (1) 809 334-1026

Emergency telephone number

CHEMTREC 1-703-527-3887
Or call 911

Other means of identification

Synonyms: Preparation based on Canthaxanthin

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Combustible Dust Combustible Dust (1) Combustible Dust

Label elements

Signal Word:
Warning

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Hazard Statement:

May form combustible dust concentration in air.

Hazards not otherwise classified

The product is under certain conditions capable of dust explosion. The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

<u>CAS Number</u>	<u>Weight %</u>	<u>Chemical name</u>
57-50-1	30.0 - 50.0%	Sucrose
91-53-2	1.0 - 5.0%	Ethoxyquin
8001-21-6	1.0 - 5.0%	Sunflower oil

4. First-Aid Measures

Description of first aid measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air.

If on skin:

Wash thoroughly with soap and water.

If in eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

If swallowed:

Rinse mouth and then drink plenty of water.

Most important symptoms and effects, both acute and delayed

Symptoms: No significant symptoms are expected due to the non-classification of the product.

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment: Symptomatic treatment (decontamination, vital functions).

5. Fire-Fighting Measures

Extinguishing media

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Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Hazards during fire-fighting:

carbon oxides

Burning produces harmful and toxic fumes. Dust explosion hazard.

Dust explosion hazard.

Advice for fire-fighters

Further information:

Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

Dusty conditions may ignite explosively in the presence of an ignition source causing flash fire.

6. Accidental release measures

Further accidental release measures:

Avoid dispersal of dust in the air (i.e., clearing dust surfaces with compressed air). Avoid the formation and build-up of dust - danger of dust explosion. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

Personal precautions, protective equipment and emergency procedures

Avoid dust formation. Information regarding personal protective measures see, section 8.

Environmental precautions

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable appliance and dispose of.

For large amounts: Contain with dust binding material and dispose of.

Dispose of absorbed material in accordance with regulations.

Nonsparking tools should be used.

7. Handling and Storage

Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

Avoid whirling up the material/product because of the danger of dust explosion. Avoid dust formation. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition. Routine housekeeping should be instituted to ensure that dusts do not accumulate on surfaces. Dry powders can build static electricity charges when subjected to the friction of transfer and mixing operations. Provide adequate precautions, such as electrical grounding and bonding, or inert atmospheres. Refer to NFPA 654, Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids (2013 Edition) for safe handling.

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Conditions for safe storage, including any incompatibilities

Suitable materials for containers: Low density polyethylene (LDPE), glass, Paper/Fibreboard, High density polyethylene (HDPE), Aluminium, tinned carbon steel (Tinplate)

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure Controls/Personal Protection

No occupational exposure limits known.

Advice on system design:

Ensure adequate ventilation. It is recommended that all dust control equipment such as local exhaust ventilation and material transport systems involved in handling of this product contain explosion relief vents or an explosion suppression system or an oxygen deficient environment. Ensure that dust-handling systems (such as exhaust ducts, dust collectors, vessels, and processing equipment) are designed in a manner to prevent the escape of dust into the work area (i.e., there is no leakage from the equipment). Use only appropriately classified electrical equipment and powered industrial trucks.

Personal protective equipment

Respiratory protection:

Wear a NIOSH-certified (or equivalent) particulate respirator. Breathing protection if breathable aerosols/dust are formed.

Hand protection:

Chemical resistant protective gloves

Eye protection:

Tightly fitting safety goggles (chemical goggles).

Body protection:

Body protection must be chosen based on level of activity and exposure.

General safety and hygiene measures:

Avoid inhalation of dust. Avoid contact with the skin, eyes and clothing. Ensure adequate ventilation. Wear protective clothing as necessary to minimize contact. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	powder
Odour:	faint specific odour
Odour threshold:	not determined
Colour:	red to brown
pH value:	6 (5 %(m), 20 °C)
Melting point:	> 100 °C
Boiling point:	not applicable
Flash point:	not applicable, the product is a solid
Flammability:	not highly flammable
Lower explosion limit:	For solids not relevant for classification and labelling.

(VDI 2263, sheet 1,
1.1)

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Upper explosion limit:	For solids not relevant for classification and labelling.
SADT:	> 75 °C Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4)
Vapour pressure:	negligible
Bulk density:	approx. 650 kg/m ³
Vapour density:	not applicable
Partitioning coefficient n-octanol/water (log Pow):	not applicable for mixtures
<i>Information on: canthaxanthin</i>	
Partitioning coefficient n-octanol/water (log Pow):	14.1 (calculated) (25 °C)
<i>Information on: Ethoxyquin</i>	
Partitioning coefficient n-octanol/water (log Pow):	3.87 (25 °C)

Thermal decomposition:	>= 150 °C
Viscosity, dynamic:	not applicable, the product is a solid
Viscosity, kinematic:	not applicable, the product is a solid
Solubility in water:	(> 35 °C) dispersible
Miscibility with water:	miscible
Evaporation rate:	negligible

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:

Corrosive effects to metal are not anticipated.

Oxidizing properties:

Based on its structural properties the product is not classified as oxidizing.

Minimum ignition energy:

> 1 J (DIN EN 13821)

The product is capable of dust explosion.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

Dust explosion hazard.

Conditions to avoid

See MSDS section 7 - Handling and storage. Avoid dust formation. Avoid all sources of ignition: heat, sparks, open flame.

Incompatible materials

None known during use and storage if used according to instructions.

Hazardous decomposition products

Decomposition products:

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Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:
≥ 150 °C

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity

Assessment of acute toxicity: Virtually nontoxic after a single ingestion.

Information on: canthaxanthin

Assessment of acute toxicity: Virtually nontoxic after a single ingestion.

Information on: Ethoxyquin

Assessment of acute toxicity: Of moderate toxicity after single ingestion. Virtually nontoxic after a single skin contact.

Oral

Information on: canthaxanthin

Type of value: LD50

Species: rat

Value: > 5,600 mg/kg (BASF-Test)

Information on: Ethoxyquin

Type of value: LD50

Species: rat

Value: 800 - 1,000 mg/kg (other)

Irritation / corrosion

Assessment of irritating effects: Not irritating to eyes and skin.

Sensitization

Assessment of sensitization: There is no evidence of a skin-sensitizing potential.

Information on: canthaxanthin

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Information on: Ethoxyquin

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Aspiration Hazard

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No data available.

Chronic Toxicity/Effects

Repeated dose toxicity

Assessment of repeated dose toxicity: No data available.

Genetic toxicity

Assessment of mutagenicity: Based on available Data, the classification criteria are not met.

Information on: canthaxanthin

Assessment of mutagenicity: No mutagenic effect was found in various tests with bacteria and mammalian cell culture. Literature data.

Information on: Ethoxyquin

Assessment of mutagenicity: The substance was mutagenic in a bacterial test system. The substance was not mutagenic in a test with mammals. Literature data.

Carcinogenicity

Assessment of carcinogenicity: Based on the ingredients there is no suspicion of a carcinogenic effect in humans.

Information on: canthaxanthin

Assessment of carcinogenicity: In long-term studies in rats and mice in which the substance was given by feed, a carcinogenic effect was not observed. Literature data.

Reproductive toxicity

Assessment of reproduction toxicity: Based on available Data, the classification criteria are not met.

Information on: canthaxanthin

Assessment of reproduction toxicity: The results of animal studies gave no indication of a fertility impairing effect. Literature data.

Information on: Ethoxyquin

Assessment of reproduction toxicity: On the basis of animal study findings, an effect on fertility cannot be excluded when given in high doses. Literature data.

Teratogenicity

Information on: canthaxanthin

Assessment of teratogenicity: No indications of a developmental toxic / teratogenic effect were seen in animal studies. Literature data.

Information on: Ethoxyquin

Assessment of teratogenicity: No indications of a developmental toxic / teratogenic effect were seen in animal studies. Literature data.

Symptoms of Exposure

No significant symptoms are expected due to the non-classification of the product.

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12. Ecological Information

Toxicity

Aquatic toxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Aquatic toxicity

Information on: Ethoxyquin

Assessment of aquatic toxicity:

Acutely toxic for aquatic organisms. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.

Toxicity to fish

Information on: canthaxanthin

LC50 (96 h) > 10,000 mg/l, Leuciscus idus (DIN 38412 Part 15, static)

The details of the toxic effect relate to the nominal concentration.

Information on: Ethoxyquin

LC50 (96 h) 18 mg/l, Oncorhynchus mykiss (OPP 72-1 (EPA-Guideline), Flow through.)

The details of the toxic effect relate to the nominal concentration.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

Information on: canthaxanthin

DIN 38412 Part 27 (draft) bacterium/EC10 (30 min): > 10,000 mg/l

The details of the toxic effect relate to the nominal concentration.

DIN EN ISO 8192-OECD 209-88/302/EEC, P. C aerobic

activated sludge, domestic/EC20 (30 min): > 1,000 mg/l

Information on: Ethoxyquin

DIN EN ISO 8192 aerobic

activated sludge, domestic/EC20 (30 min): approx. 60 mg/l

Persistence and degradability

Assessment biodegradation and elimination (H2O)

The product has not been tested.

Assessment biodegradation and elimination (H2O)

Information on: canthaxanthin

Not readily biodegradable (by OECD criteria). Poorly biodegradable.

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Information on: Ethoxyquin

Not readily biodegradable (by OECD criteria). Poorly biodegradable.

Elimination information

Information on: canthaxanthin

< 20 % BOD of the ThOD (28 d) (OECD 301F; ISO 9408; 92/69/EEC, C.4-D) (aerobic, activated sludge, domestic)

Information on: Ethoxyquin

< 20 % BOD of the ThOD (25 d) (OECD Guideline 301 F) (aerobic, activated sludge, industrial)

Bioaccumulative potential

Assessment bioaccumulation potential

The product has not been tested.

Assessment bioaccumulation potential

Information on: Ethoxyquin

No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).

Information on: canthaxanthin

The product will not be readily bioavailable due to its consistency and insolubility in water. No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).

Mobility in soil

Assessment transport between environmental compartments

not determined

13. Disposal considerations

Waste disposal of substance:

Observe national and local legal requirements.

Dispose of in accordance with national, state and local regulations.

Container disposal:

Dispose of in accordance with national, state and local regulations.

14. Transport Information

Land transport

TDG

Hazard class:

4.2

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Packing group: III
ID number: UN 3088
Hazard label: 4.2
Proper shipping name: SELF-HEATING SOLID, ORGANIC, N.O.S. (contains CANTHAXANTHIN)

Sea transport

IMDG

Hazard class: 4.2
Packing group: III
ID number: UN 3088
Hazard label: 4.2
Marine pollutant: NO
Proper shipping name: SELF-HEATING SOLID, ORGANIC, N.O.S. (contains CANTHAXANTHIN)

Air transport

IATA/ICAO

Hazard class: 4.2
Packing group: III
ID number: UN 3088
Hazard label: 4.2
Proper shipping name: SELF-HEATING SOLID, ORGANIC, N.O.S. (contains CANTHAXANTHIN)

Further information

Not dangerous goods of class 4.2 in packages up to 3000 litres capacity.

15. Regulatory Information

Federal Regulations

Not applicable

NFPA Hazard codes:

Health: 0 Fire: 1 Reactivity: 0 Special:

16. Other Information

SDS Prepared by:

BASF NA Product Regulations

SDS Prepared on: 2017/11/22

We support worldwide Responsible Care® initiatives. We value the health and safety of our employees, customers, suppliers and neighbors, and the protection of the environment. Our commitment to Responsible Care is integral to conducting our business and operating our facilities in a safe and environmentally responsible fashion, supporting our customers and suppliers in ensuring the safe and environmentally sound handling of our products, and minimizing the impact of our operations on society and the environment during production, storage, transport, use and disposal of our products.

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This information is considered accurate but is not exhaustive and shall only be used as a guideline based on current knowledge of the chemical substance or mixture. Safety precautions suitable for the product must be applied.

END OF DATA SHEET